

Day4: Connecting and Testing Peripherals Practical (1 hors):
Set up and testing peripherals like printers and webcams.
Task: resolve connectivity issues for a given setup.

Task 1: Connect and Test a USB Webcam

Aim

To connect a USB webcam to the computer, install the required driver, and verify that the webcam is working properly.

Procedure

Step 1: I connected the USB webcam to an available USB port on the computer.

Step 2: Windows automatically detected the webcam. I waited for Windows to install the required driver.

Step 3: I opened the **Camera** application from the Windows Start menu.

Step 4: The webcam was selected automatically and the live camera preview was displayed on the screen.

Step 5: I checked the camera preview to make sure that the webcam was working properly.

Step 6: I clicked the **Take Photo** button and captured a photo using the webcam.

Step 7: I checked the captured photo to confirm that the webcam was working correctly.

Result

The USB webcam was successfully connected and detected by the computer. The webcam worked correctly in the Windows Camera application and a photo was successfully captured.

Task 2: Set Up a Local Printer and Print a Test Page

Aim

To connect a printer, install the correct driver, and print a test page.

Procedure

Step 1: I connected the printer to the computer using a USB cable.

Step 2: I switched on the printer and waited for Windows to detect the device.

Step 3: I opened: **Settings** → **Bluetooth & devices** → **Printers & scanners**

Step 4: I clicked **Add device** and waited for Windows to search for available printers.

Step 5: The connected printer was displayed in the list.

Step 6: I selected the printer and allowed Windows to install the required printer driver.

Step 7: After installation, I opened the printer settings and checked that the printer status was **Ready**.

Step 8: I selected the option to print a **test page**.

Step 9: The printer printed the test page successfully.

Result

The printer was successfully connected and installed. The test page was printed successfully, confirming that the printer was working properly.

Task 3: Troubleshoot “Printer Not Detected” Problem

Aim

To simulate a printer connectivity problem and resolve it using Windows Device Manager.

Procedure

Step 1: I disconnected the printer's USB cable from the computer to simulate a connectivity problem.

Step 2: I checked the printer from Windows and found that the printer was not detected or available.

Step 3: I opened **Device Manager** from the Windows Start menu.

Step 4: I checked the device categories such as **Printers**, **Print queues**, and **Universal Serial Bus controllers**.

Step 5: I looked for any missing device or error icon that could indicate a hardware or driver problem.

Step 6: I reconnected the printer's USB cable firmly to the computer.

Step 7: In Device Manager, I opened the **Action** menu.

Step 8: I selected **Scan for hardware changes**.

Step 9: Windows scanned the computer and detected the connected printer again.

Step 10: I checked the printer in **Settings → Bluetooth & devices → Printers & scanners**.

Result

The printer was successfully detected again after reconnecting the USB cable and using the **Scan for hardware changes** option in Device Manager.

Task 4: Troubleshoot Webcam Not Working in Zoom

Problem

The webcam was working correctly in the Windows Camera application, but the webcam video was not displaying in Zoom.

Aim : To troubleshoot and resolve the webcam issue in Zoom.

Procedure

Step 1: First, I opened the Windows Camera application and confirmed that the webcam was working correctly.

Step 2: I closed the Camera application because another application can sometimes use the webcam.

Step 3: I closed other applications that could be using the camera.

Step 4: I opened **Zoom**.

Step 5: I opened: **Zoom → Settings → Video**

Step 6: Under the **Camera** option, I checked the available cameras.

Step 7: I selected the correct **USB Webcam** from the camera list.

Step 8: I checked Windows camera permissions by opening:

Settings → Privacy & security → Camera

Step 9: I made sure that **Camera access** was enabled and that desktop applications were allowed to access the camera.

Step 10: I returned to Zoom and checked the video preview.

Step 11: The webcam video appeared correctly in Zoom.

Result

The webcam problem in Zoom was successfully resolved. The issue was related to the camera selection/permission and another application potentially using the webcam. After selecting the correct camera and checking permissions, the webcam worked properly in Zoom.